

# CS310 Natural Language Processing

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## Assignment 1 (written)

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**Total points: 10**

Consider a softmax layer of output size  $k$  ( $k > 1$ ). Within unit  $i$  ( $i = 1, \dots, k$ ), the linear before activation is  $z_i$  (also called logits), and the output after activation is  $o_i$ . We have the following equation for forward propagation:

$$o_i = \frac{\exp(z_i)}{\sum_{j=1}^k \exp(z_j)}$$

The cross-entropy loss between an output prediction  $\hat{y} = [o_1, \dots, o_k]$  and the ground truth label  $y = [y_1, \dots, y_k]$  (one-hot encoded) is defined as:

$$L_{CE} = - \sum_{i=1}^k y_i \log(\hat{y}_i) = - \sum_{i=1}^k y_i \log o_i$$

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### Question 1

Show that when  $k = 2$ , the loss is equivalent to binary cross-entropy (BCE) loss

$$L_{BCE} = -y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i).$$

(4 points, show your math derivation to get the credits)

**【Answer】**

For  $k = 2$ , the output prediction is  $\hat{y} = [o_1, o_2]$  and the ground truth label is  $y = [y_1, y_2]$ .

Because  $y$  is one-hot encoded, we have  $y_1 + y_2 = 1$ , which means  $y_2 = 1 - y_1$ .

Because of the softmax, we have  $o_1 + o_2 = 1$ , which means  $o_2 = 1 - o_1$ .

The cross-entropy loss for  $k = 2$  is:

$$L_{CE} = -(y_1 \log o_1 + y_2 \log o_2)$$

Substituting  $y_2 = 1 - y_1$  and  $o_2 = 1 - o_1$ :

$$L_{CE} = -(y_1 \log o_1 + (1 - y_1) \log(1 - o_1))$$

$$L_{CE} = -y_1 \log o_1 - (1 - y_1) \log(1 - o_1)$$

If we let  $y_i$  denote the ground truth for class 1 and  $\hat{y}_i$  denote the predicted probability  $o_1$ :

$$L = -y_i \log \hat{y}_i - (1 - y_i) \log(1 - \hat{y}_i)$$

This is BCE loss formula.

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### Question 2

What is the gradient w.r.t the  $i$ -th unit  $\frac{\partial L}{\partial z_i}$ ?

(6 points, show your math derivation to get the credits)

**【Answer】**

The loss function is  $L = - \sum_{j=1}^k y_j \log o_j$ .

First, compute  $\frac{\partial L}{\partial o_j}$ :

$$\frac{\partial L}{\partial o_j} = -\frac{y_j}{o_j}$$

Next, compute the derivative of the softmax function  $\frac{\partial o_j}{\partial z_i}$ :

- If  $j = i$ :
$$\frac{\partial o_i}{\partial z_i} = \frac{\exp(z_i) \sum_m \exp(z_m) - \exp(z_i) \exp(z_i)}{(\sum_m \exp(z_m))^2} = o_i(1 - o_i)$$

- If  $j \neq i$ :  

$$\frac{\partial o_j}{\partial z_i} = \frac{0 \cdot \sum_m \exp(z_m) - \exp(z_j) \exp(z_i)}{(\sum_m \exp(z_m))^2} = -o_j o_i$$

substitute these back into the gradient equation:

$$\begin{aligned}\frac{\partial L}{\partial z_i} &= \frac{\partial L}{\partial o_i} \frac{\partial o_i}{\partial z_i} + \sum_{j \neq i} \frac{\partial L}{\partial o_j} \frac{\partial o_j}{\partial z_i} \\ \frac{\partial L}{\partial z_i} &= -\frac{y_i}{o_i} [o_i(1 - o_i)] + \sum_{j \neq i} \left(-\frac{y_j}{o_j}\right) (-o_j o_i) \\ \frac{\partial L}{\partial z_i} &= -y_i(1 - o_i) + \sum_{j \neq i} y_j o_i \\ \frac{\partial L}{\partial z_i} &= -y_i + y_i o_i + o_i \sum_{j \neq i} y_j\end{aligned}$$

Since it is one-hot encoded,  $\sum_{j=1}^k y_j = 1$ , we have  $\sum_{j \neq i} y_j = 1 - y_i$ .

$$\begin{aligned}\frac{\partial L}{\partial z_i} &= -y_i + y_i o_i + o_i(1 - y_i) \\ \frac{\partial L}{\partial z_i} &= -y_i + y_i o_i + o_i - y_i o_i \\ \frac{\partial L}{\partial z_i} &= o_i - y_i\end{aligned}$$

Therefore, the gradient is:

$$\frac{\partial L}{\partial z_i} = o_i - y_i$$